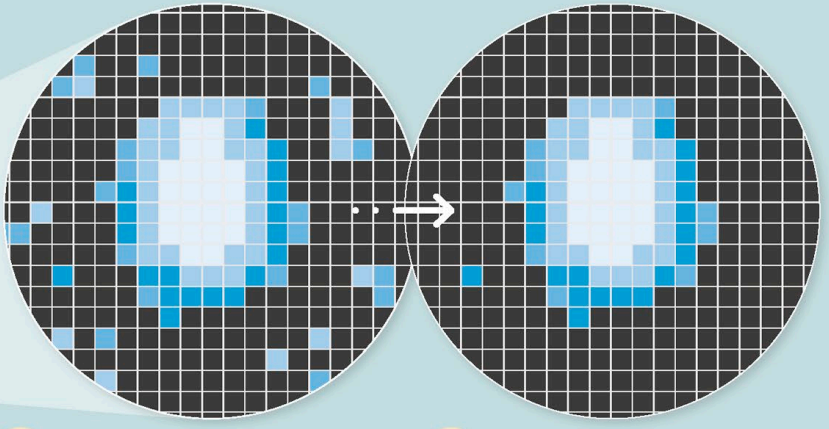
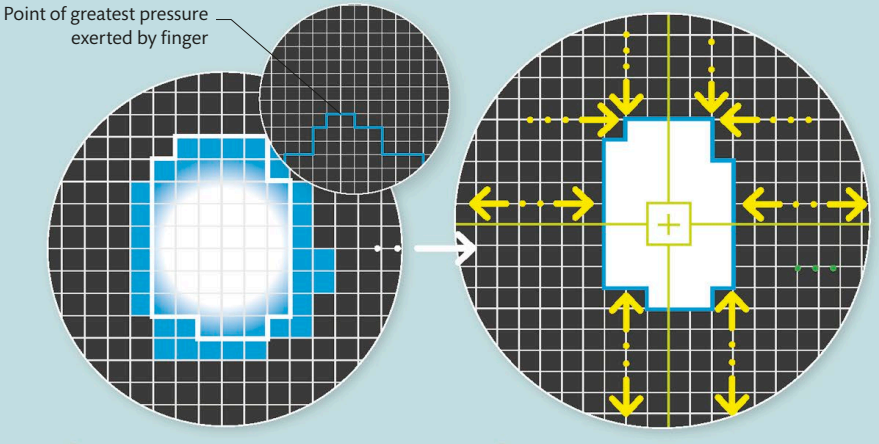


2 Raw data captured
Measurements of changes in the electric current are taken at every point in the grid. The points directly beneath the fingertip experience the largest drop.

3 Noise removed
Electromagnetic interference, or noise, must be filtered out to ensure a strong and stable touch response. This noise can come from external sources such as chargers.



4 Pressure points measured
The size and shape of the areas of the grid in contact with the user's fingertips are identified to determine the points where the greatest pressure was applied.

5 Exact coordinates calculated
Electric signals from each point on the grid are sent to the device's main processor, which uses the data to calculate the precise position of the fingertip.

CONNECTIVITY

One of the most useful features of mobile devices is their ability to connect and communicate with other devices nearby. The devices could be physically connected, but it is usually more convenient to exchange data wirelessly using radio signals.

Bluetooth
Bluetooth uses radio waves to communicate over short distances. It allows wireless connectivity to other devices using radio signals, including Bluetooth headsets.

Wi-Fi
Wi-Fi (see pp.202–203) allows local networks of devices to communicate wirelessly via a router, which also connects to the Internet.

RFID
Radio-frequency ID tags—often attached to objects in stores or factories—emit unique radio waves, from which they can be identified by mobile devices.

NFC
Near-field communication allows two very close devices to communicate. This is used in contactless payment systems and keycards.



Smartwatch
These miniaturized computers feature many of the functions of a smartphone.



Gaming platform
Some gaming systems contain the screen, controls, speakers, and console within a single device.



e-Reader
e-Readers are designed for reading electronic books. Many use electronic paper (see pp.208–209).



PDA
Personal digital assistants (PDAs) are information managers. Most can access the Internet and work as a phone.